# COUPLING, RECORDS, AND STABLE RELATIONS: FORMATION, PROTECTED CONTENT AND THE LIFETIME OF A GEOMETRY

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**ABSTRACT**

A coupling can preserve an inherited relation without having formed it. We ask how interactions select stable structure and how records make that structure usable. Commuting Hermitian noise selects a direct-sum algebra on a given carrier; its central sectors do not require an independently chosen spatial basis. A separated range of decay rates further selects a unique block partition over a finite window within the stated block class. We distinguish this selection from the formation of correlations and from learning the operations that use them. A collective recorder creates a flagged protected Bell branch from a product input, while a physical link creates a deterministic protected Bell pair across noise-disconnected components. Both retain the resources that produce the relation. Block-constant records preserve internal quantum content, whereas independent leakage imposes a sharp $R^{-1/2}$ symmetry threshold for $R$ records. An exact entropy identity separates sector and within-sector contributions to a supplied correlation geometry. The GHZ example then exhibits a selected geometry, its finite plateau and its loss under symmetry breaking. Finite experimental capabilities remain continuous under perturbations even when the exact fixed algebra changes discontinuously. These results support a process-selected subsystem programme: an effective component must acquire reproducible operations as well as stable correlations. The carrier, available interactions and spatial interpretation remain distinct inputs to be tested rather than inferred from a noise graph alone.

## 1 Introduction

A stable distinction is a physical achievement. It must survive the interactions that would erase it, be readable through an available instrument, and last long enough to matter. An algebra of invariant observables describes one part of that achievement; it does not specify a reader, a preparation history or a set of available gates.

Paper A [1] shows why the distinction matters for geometry. Channels generating the same observable algebra can resolve different states and produce non-isometric finite correlation geometries. Its local bit-flip family has a Stinespring realization and a finite fragment-readout certificate. Here we ask a complementary question: what structure a coupling protects, how actual records and interactions form usable relations within that structure, and when those relations lose their support.

Commuting Hermitian noise gives an exact starting point. The fixed observables are the commutant of the coupling operators, familiar from einselection and noiseless codes [9, 10, 12, 13]. The infinite-time channel pinches onto this algebra. At finite times its operator range is full, so a protected block algebra is additional stability data rather than the range of the actual channel. A window and a tolerance give a unique block partition when intra-block and inter-block rates are separated (Theorem 4.2). The maximality statement concerns block algebras in the specified joint eigenbasis, not every rotated algebra or every useful finite task.

Recording adds a different restriction. A coupling constant on each block can record its label without learning its quantum content. Independent fragments that resolve two states within a block destroy their coherence according to a multiplicative visibility budget (Theorem 5.2). The centre and the blocks thus perform different jobs within one stable structure. They are not disjoint sources of geometry: classical sector weights can sustain mutual information even when every block is one-dimensional. Section 6 gives the exact decomposition needed to distinguish the contributions.

Selection alone also says too little about formation. Two disconnected collective-noise components can preserve a logical Bell pair shared at the start. Conversely, local independent processes cannot create cross-component correlations from independent inputs. Section 7 supplies the missing process distinction and two explicit formation mechanisms. One retains a collective measurement flag and all its branches; the other uses an actual cross-component Hamiltonian link. Both generate relations from product inputs, and both expose the interaction resources responsible for them. Foundation III [6] supplies a further mechanism in which acquired records select an encoder and decoder for the same pre-existing quantum content. Storage, formation and learned use then have separate, testable meanings.

The GHZ calculation in Section 8 shows how a coupling changes a finite correlation geometry and how weak symmetry breaking ends its plateau. For noncommuting perturbations, Section 4.1 bounds complete adaptive record laws and task risks without requiring a stable algebra. A transverse qubit exhibits a diverging record lifetime despite a scalar exact fixed algebra.

The Hilbert space and coupling permissions are supplied. The formation and GHZ experiments additionally specify site factors; the exact sector-selection result does not need a prior spatial factorization. Consciousness I [4] selects a Virasoro fixed net inside a given current-net representation; Foundation's hulls concern closure under declared operations. Neither result derives the ambient carrier. Our formation results address how relations and usable protected sectors arise on that carrier. Section 2 states the effective-subsystem question before the constructions. Section 10 examines its extension beyond commuting noise.

## 2 What has to form

The generation question has three physically different outputs. A coupling can select a direct-sum sector without supplying a tensor factorization. An interaction can create a relation between supplied components without teaching a reader how to use it. Acquiring a component also requires repeatable operations whose errors stay controlled over the intended task. These distinctions set the target of the constructions below.

**Conjecture 2.1** (Process-selected effective subsystems)**.** *On an initially specified carrier, robust effective subsystems form through a history of interactions that both builds relations and makes a noncommuting set of internal operations reproducible across a finite task window. The relevant connectivity is that of the complete physical process, including recording ports and Hamiltonian controls, not the collective-noise hypergraph alone. Within a separated-rate regime, a learned encoding can turn the process's indistinguishable sectors into stable operational components; their boundaries change when the reproducible operations or their lifetime change.*

The proposal is tested by changing the preparation history, recording port or coupling while keeping the task fixed. A stable Bell pair present initially demonstrates preservation. A product preparation followed by the actual collective port or Hamiltonian of Section 7 demonstrates formation of a relation. A diagnostic record that selects a working encoder demonstrates learned use. None of these tests alone establishes a complete subsystem gate algebra. This is why the complete interaction history, rather than just the noise graph or the final state, is the candidate source of effective organization.

The mathematical starting point needs only a carrier and its specified couplings. Site factors are introduced later for correlation and formation experiments. The general tensor-factorization question is therefore kept separate from the exact selection of stable sectors.

## 3 Reductions selected by a coupling

### 3.1 Setting

Let $\mathcal{H}$ be a finite-dimensional system Hilbert space. The environment couples to the system through Hermitian operators $A_1, \ldots, A_K$ with $[A_k, A_l] = 0$, at rates $\gamma_k > 0$. In the Markovian limit the reduced dynamics is generated by

$$\mathcal{L}(\rho) = \sum_{k=1}^{K} \gamma_k \Big( A_k \rho A_k - \tfrac{1}{2}\{A_k^2, \rho\} \Big), \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

A system Hamiltonian that commutes with every $A_k$ only rotates states inside the blocks defined below; we work in the corresponding interaction picture and return to Hamiltonians that do not commute in Remark 3.3. Section 5 treats the same couplings in an exact unitary model with environmental fragments.

Choose an orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$ of joint eigenvectors, $A_k|x\rangle = a_k(x)|x\rangle$, and put

$$D(x,y) = \tfrac{1}{2}\sum_k \gamma_k\big(a_k(x) - a_k(y)\big)^2 = \|u(x) - u(y)\|^2, \qquad u(x) = \big(\sqrt{\gamma_k/2}\, a_k(x)\big)_{k=1}^{K} \in \mathbb{R}^K. \tag{2}$$

We call $u(\mathcal{X}) \subset \mathbb{R}^K$ the rate image of the basis; decoherence rates are squared Euclidean distances in it. Write $x \approx y$ when $a_k(x) = a_k(y)$ for every $k$, let $\pi_\infty$ be the partition of $\mathcal{X}$ into $\approx$-classes $a$, and let $P_a$ be the projection onto $\mathcal{H}_a = \operatorname{span}\{|x\rangle : x \in a\}$. For any partition $\pi$ of $\mathcal{X}$ into classes $C$, put

$$\mathfrak{B}_\pi = \bigoplus_{C\in\pi} \mathcal{B}(\mathcal{H}_C), \qquad \mathbb{E}_\pi(\rho) = \sum_{C\in\pi} P_C \rho P_C.$$

$\mathfrak{B}_\pi$ is a unital $*$-algebra, $\mathbb{E}_\pi$ is a channel (the pinching), and $\mathbb{E}_\pi$ is its own dual, the conditional expectation onto $\mathfrak{B}_\pi$.

**Proposition 3.1** (The reduction selected by a commuting coupling)**.** *1. $\Phi_t$ acts as a Schur multiplier in the eigenbasis: $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$, and $\Phi_t^* = \Phi_t$.*

2. *The observables fixed by every $\Phi_t^*$ form the algebra $\mathfrak{B}_{\pi_\infty} = \{A_1, \dots, A_K\}'$.*
3. *$\Phi_t \to \mathbb{E}_{\pi_\infty}$ as $t \to \infty$, and the algebra generated by the dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$. In the terminology of Paper A, $(\mathfrak{B}_{\pi_\infty}, \mathbb{E}_{\pi_\infty})$ is a minimally compatible coarse-graining structure.*
4. *For every finite $t$, $\Phi_t$ is invertible on the operator space, so the algebra generated by its dual range is $\mathcal{B}(\mathcal{H})$.*

*Proof.* (1) In the eigenbasis, $(A\rho A - \frac{1}{2}\{A^2, \rho\})_{xy} = (a(x)a(y) - \frac{1}{2}a(x)^2 - \frac{1}{2}a(y)^2)\rho_{xy} = -\frac{1}{2}(a(x) - a(y))^2\rho_{xy}$, so $\mathcal{L}$ multiplies the $(x, y)$ entry by $-D(x, y)$. The multiplier is real and symmetric, hence $\Phi_t^* = \Phi_t$. (2) $X$ is fixed for all $t$ exactly when $X_{xy} = 0$ whenever $D(x, y) > 0$, that is, whenever $x \not\approx y$. An operator commutes with every diagonal $A_k$ exactly when it has no entries between basis vectors with different eigenvalue tuples. (3) Entries with $D(x, y) > 0$ tend to zero and the others are constant. The dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$, which is already an algebra. (4) All multipliers $e^{-tD(x,y)}$ are nonzero. □

Part (2) is the elementary commuting case of the known structure of fixed points of quantum dynamical semigroups [14, 15]. For this series the consequence matters more than the statement: the given coupling determines both its channel and its invariant algebra. Part (4) is Paper A's result that an algebra label does not record resolution, now in dynamical form. Before the long-time limit the channel generates everything, and the structure that a coupling makes accessible has to be identified by how well observables survive over time.

**Corollary 3.2** (The selected sectors are intrinsic)**.** *In the interaction picture of* (1)*, the minimal central projections of* $\operatorname{Fix}(\Phi)$ *are precisely the complete joint-eigenvalue projections $P_a$. They are determined by the semigroup up to relabelling. A common unitary conjugation of the carrier and all couplings conjugates these projections and preserves the rates between them.*

*Proof.* Proposition 3.1 gives $\operatorname{Fix}(\Phi) = \bigoplus_a \mathcal{B}(P_a\mathcal{H})$. The centre of this algebra is $\operatorname{span}\{P_a\}$; its minimal nonzero projections are unique. On $P_a\mathcal{B}(\mathcal{H})P_b$, the generator is multiplication by $-\frac{1}{2}\sum_k \gamma_k(a_k(a) - a_k(b))^2$. Conjugating the entire model preserves these scalars. □

The eigenbasis used in the proof is not an extra choice of spatial sites. What has been selected is a direct sum, including quantum content within its blocks. It neither supplies the instruments for that content nor identifies the blocks with spatial tensor factors.

The objects of Paper E [3] can be located in this class. Its environmental fixed set $\operatorname{Fix}(\mathcal{E})$ is $\mathfrak{B}_{\pi_\infty}$. In Paper E's collision model, with a qubit pointer, the pointer algebra equals the fixed algebra of the induced dephasing and contains every effect a fragment can report. The centre of $\mathfrak{B}_{\pi_\infty}$ is generated by the block projections $P_a$; it is commutative, and Section 5 shows that it is what the environment records, so it serves as Paper E's pointer or record algebra. The inclusion of the record algebra in the fixed set is thus a conclusion of the model, as Paper E requires of any such overlap. The fixed set is strictly larger than its centre as soon as some block has dimension greater than one; a nontrivial two-sector qubit pointer has only one-dimensional blocks. Such blocks are decoherence-free subspaces [12, 13]. They retain quantum content; sector weights can retain classical correlations as well.

**Remark 3.3** (Hamiltonians that do not commute with the couplings)**.** *If the system Hamiltonian $H$ has a part $H_{\mathrm{off}}$ connecting different blocks, $\mathfrak{B}_{\pi_\infty}$ need not remain exactly invariant. Strong dephasing can then confine the dynamics to the blocks, with leakage on a time scale of order $\Gamma_{\mathrm{out}}/\|H_{\mathrm{off}}\|^2$, where $\Gamma_{\mathrm{out}}$ is the smallest between-block rate; this is the mechanism of quantum Zeno subspaces [16]. Section 4.1 gives an exact two-level lifetime and a finite-time stability bound without a commutativity assumption. Paper D's chain with a transverse field requires its own many-body estimate.*

## 4 Selection over a finite window

**Definition 4.1** (Stable algebra on a window)**.** *A unital $*$-subalgebra $\mathfrak{B} \subseteq \mathcal{B}(\mathcal{H})$ is $\varepsilon$-stable on $[0,T]$ if*

$$\sup_{0\le t\le T}\ \sup_{X\in\mathfrak{B},\ \|X\|\le 1} \|\Phi_t^*(X) - X\| \le \varepsilon.$$

The diagnostic tests the whole unit ball of an algebra, rather than selected states or a list of slow observables. For a partition $\pi$ define the width of a class $C$ in the direction $k$, $w_k(C) = \max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)$, and the two rates

$$\Gamma_{\mathrm{in}}(\pi) = \max_{C\in\pi}\ \tfrac{1}{2}\sum_k \gamma_k w_k(C)^2, \qquad \Gamma_{\mathrm{out}}(\pi) = \min\{D(x,y) :\ x,y \text{ in different classes of } \pi\}. \tag{3}$$

The minimum over an empty set is $+\infty$, so a single-block partition has $\Gamma_{\mathrm{out}} = +\infty$. If every coupling is scalar, the full algebra is fixed and selected for every window. In the rate image, $\Gamma_{\mathrm{in}}(\pi)$ is the largest squared diagonal of the axis-parallel boxes enclosing the classes, and $\Gamma_{\mathrm{out}}(\pi)$ is the smallest squared distance between points of different classes.

**Theorem 4.2** (Selection by a window)**.** *Let $0 < \varepsilon < 1$ and $T > 0$.*

1. *For every $X \in \mathfrak{B}_\pi$ and $t \ge 0$, $\|\Phi_t^*(X) - X\| \le t\,\Gamma_{\mathrm{in}}(\pi)\|X\|$. For every state $\rho \in \mathfrak{B}_\pi$, $\|\Phi_t(\rho) - \rho\|_1 \le t\,\Gamma_{\mathrm{in}}(\pi)$. In particular $\mathfrak{B}_\pi$ is $\varepsilon$-stable on $[0,T]$ when $T\Gamma_{\mathrm{in}}(\pi) \le \varepsilon$.*
2. *If $1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$, no unital $*$-algebra that contains $\mathfrak{B}_\pi$ properly is $\varepsilon$-stable on $[0,T]$.*
3. *Call $\pi$ selected by $(T,\varepsilon)$ if*
$$T\,\Gamma_{\mathrm{in}}(\pi) \le \varepsilon < 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)}. \tag{4}$$
*At most one partition is selected by $(T,\varepsilon)$. If $\pi$ is selected by $(T,\varepsilon)$ and $\pi'$ by $(T',\varepsilon)$ with $T \le T'$, then $\pi'$ refines $\pi$, so $\mathfrak{B}_{\pi'} \subseteq \mathfrak{B}_\pi$.*
4. *For $T > -\ln(1-\varepsilon)/\Gamma_{\mathrm{out}}(\pi_\infty)$ the selected partition exists and is $\pi_\infty$, so the selected algebra is the fixed algebra of Proposition 3.1.*

*Proof.* (1) $\mathfrak{B}_\pi$ is invariant under $\mathcal{L}$, and on it $\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k,[A_k,X]]$. On a class $C$, replacing $A_k$ by $A_k - m_k$ with $m_k$ the midpoint of its eigenvalues on $C$ does not change the commutator and gives an operator of norm $w_k(C)/2$; hence $\|[A_k,X_C]\| \le w_k(C)\|X_C\|$ and $\|[A_k,[A_k,X_C]]\| \le w_k(C)^2\|X_C\|$. Since $\|X\| = \max_C \|X_C\|$, $\|\mathcal{L}(X)\| \le \Gamma_{\mathrm{in}}(\pi)\|X\|$. Then $\Phi_t^*(X) - X = \int_0^t \Phi_s^*(\mathcal{L}(X))\,ds$ and $\Phi_s^*$ is a contraction in operator norm. For a state $\rho \in \mathfrak{B}_\pi$ and $\|Y\| \le 1$, $\mathrm{Tr}[(\Phi_t(\rho) - \rho)Y] = \mathrm{Tr}[\rho(\Phi_t^* - \mathrm{id})(\mathbb{E}_\pi Y)]$, because $\rho = \mathbb{E}_\pi\rho$, $\mathbb{E}_\pi$ is self-dual and commutes with the Schur multiplier $\Phi_t^*$; since $\mathbb{E}_\pi Y \in \mathfrak{B}_\pi$ and $\|\mathbb{E}_\pi Y\| \le 1$, the first statement bounds this by $t\Gamma_{\mathrm{in}}(\pi)$.

(2) Let $\mathfrak{B} \supsetneq \mathfrak{B}_\pi$ and $X \in \mathfrak{B} \setminus \mathfrak{B}_\pi$. Some entry $X_{x'y'}$ with $x' \in C$, $y' \in C'$, $C \ne C'$, is nonzero. For any $x \in C$, $y \in C'$ the matrix units $E_{xx'}$ and $E_{y'y}$ lie in $\mathfrak{B}_\pi \subseteq \mathfrak{B}$, so $E_{xx'}XE_{y'y} = X_{x'y'}E_{xy} \in \mathfrak{B}$. Then $\|\Phi_T^*(E_{xy}) - E_{xy}\| = 1 - e^{-TD(x,y)} \ge 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$.

(3) Let $\pi$ and $\pi'$ be selected by $(T,\varepsilon)$ and $(T',\varepsilon)$, $T \le T'$. If $x,y$ lie in one class of $\pi'$, then $D(x,y) \le \Gamma_{\mathrm{in}}(\pi') \le \varepsilon/T' \le \varepsilon/T$. If $x,y$ lay in different classes of $\pi$, then $D(x,y) \ge \Gamma_{\mathrm{out}}(\pi) > -\ln(1-\varepsilon)/T \ge \varepsilon/T$. Hence every class of $\pi'$ lies in a class of $\pi$. For $T = T'$ the same argument with the roles exchanged gives $\pi = \pi'$.

(4) $\Gamma_{\mathrm{in}}(\pi_\infty) = 0$ and $\Gamma_{\mathrm{out}}(\pi_\infty) > 0$, so $\pi_\infty$ satisfies (4) for such $T$; uniqueness is part (3). □

A window and a tolerance therefore pick out at most one algebra of the form $\mathfrak{B}_\pi$, and they pick one out when the rate image of the basis splits into clusters that are small compared with $\varepsilon/T$ and far apart compared with $-\ln(1-\varepsilon)/T$. This is a separation of time scales, the mechanism behind metastability in open quantum systems [17]; for commuting couplings it can be stated exactly. The selected algebra is moreover the largest $\varepsilon$-stable algebra of block form: if $\mathfrak{B}_\sigma$

is $\varepsilon$-stable on $[0,T]$, every matrix unit $E_{xy}$ with $x,y$ in one class of $\sigma$ satisfies $1-e^{-TD(x,y)} \le \varepsilon$, so $x$ and $y$ lie in one class of the selected partition $\pi$, and $\mathfrak{B}_\sigma \subseteq \mathfrak{B}_\pi$. As the window lengthens, the selected algebra shrinks in discrete steps located at the gaps of the rate spectrum and ends at the fixed algebra. In the order of Paper A this is a chain of further coarse-grainings, since $\mathbb{E}_{\pi'} = \mathbb{E}_{\pi'} \circ \mathbb{E}_\pi$ when $\pi'$ refines $\pi$.

Without a gap there is no selected algebra at that window. Selection uses the sufficient bound of part (1), so between that bound and the necessary bound of part (2) a unique maximal stable block algebra can still exist; but the window may also fail to determine one, and two incomparable maximal stable block algebras can occur. The set of $\varepsilon$-stable matrix units can then be an operator system whose generated algebra contains unstable elements: if $D(x,y)$ and $D(y,z)$ are small, $D(x,z)$ can still be as large as $(\sqrt{D(x,y)}+\sqrt{D(y,z)})^2$, and a chain of small steps can connect distant points of the rate image. Theorem 4.2 is a sufficient selection certificate. Failure of its gap condition does not by itself decide existence or uniqueness of a maximal stable algebra. Nor does stability of a block algebra supply the instruments needed to use every element as a gate.

### 4.1 Continuous capabilities when the fixed algebra jumps

A weak coupling can change the exact commutant discontinuously. It cannot change every finite experiment discontinuously. The distinction matters when a selected algebra is used to describe a finite observer.

**Proposition 4.3** (Finite-time operational stability)**.** *Let $\mathcal{L}_0, \mathcal{L}_1$ be finite-dimensional Lindblad generators, and put $v = \|\mathcal{L}_1 - \mathcal{L}_0\|_\diamond$. Then*

$$\|e^{t\mathcal{L}_1} - e^{t\mathcal{L}_0}\|_\diamond \le \min\{2, tv\}. \tag{5}$$

*Consider any experiment with the same initial states, retained ancillas and permitted instruments in both models. Between its instruments the system evolves for prescribed intervals $t_1,\dots,t_m$, with total $T=\sum_i t_i$. Instruments may depend on earlier outcomes. The total variation distance between its complete record laws is at most $\min\{1, Tv/2\}$. Consequently, for a common decision problem with loss in $[0,L]$, the two optimal risks differ by at most $L\min\{1,Tv/2\}$. In particular, a Hamiltonian perturbation $-i\epsilon[K,\cdot]$ gives $v \le 2|\epsilon|\|K\|$.*

*Proof.* Duhamel's formula writes the channel difference as $\int_0^t e^{(t-s)\mathcal{L}_1}(\mathcal{L}_1-\mathcal{L}_0)e^{s\mathcal{L}_0}\,ds$. CPTP maps have diamond norm one, giving $tv$; two channels differ by at most two. A hybrid argument replaces the evolution intervals one at a time, keeping outcome-controlled instruments as CPTP maps on the system and classical record. Contractivity and the triangle inequality give a trace-norm difference at most $Tv$. The bounded-loss estimate holds uniformly over each common policy, so it also holds for the infima. Finally, $\|[K,A]\|_1 \le 2\|K\|\|A\|_1$ holds with every retained ancilla. □

The estimate needs neither a gap nor a selected algebra. It bounds distinguishing, storing and acting through the permitted experiment, not just the state of an isolated system. The available instruments and the total time remain the same on both sides. A discontinuous change of permissions, or an unbounded observation time, is a different comparison.

For an exact noncommuting example, take

$$\mathcal{L}_\epsilon(\rho) = -i\epsilon[X,\rho] + \kappa(Z\rho Z - \rho), \qquad \kappa > 0. \tag{6}$$

The fixed algebra is $\operatorname{span}\{I,Z\}$ at $\epsilon = 0$ and only $\mathbb{C}I$ at every $\epsilon \neq 0$. To see the latter, the Hilbert–Schmidt dissipative form of a fixed observable forces $[Z,O]=0$; the remaining Hamiltonian equation then forces $[X,O]=0$. Nevertheless, the two initially perfect records $(I \pm Z)/2$ remain readable for a long time. If $0 < 2|\epsilon| < \kappa$, put $\omega = \sqrt{\kappa^2 - 4\epsilon^2}$. Their Bloch vectors at time $t$ are $\pm(0, y(t), z(t))$, where

$$z(t) = e^{-\kappa t}\left(\cosh\omega t + \frac{\kappa}{\omega}\sinh\omega t\right), \qquad y(t) = -\frac{2\epsilon}{\omega}e^{-\kappa t}\sinh\omega t. \tag{7}$$

These expressions solve $\dot y = -2\kappa y - 2\epsilon z$, $\dot z = 2\epsilon y$, with $(y,z) = (0,1)$ initially. Reading only $Z$ has error $(1-z(t))/2$; allowing the optimal system measurement gives error $(1-\sqrt{y(t)^2+z(t)^2})/2$. The slow decay rate is exactly

$$r = \kappa - \sqrt{\kappa^2 - 4\epsilon^2} = \frac{2\epsilon^2}{\kappa} + O(\epsilon^4/\kappa^3). \tag{8}$$

Thus arbitrarily weak transverse coupling erases the infinite-time record, while its finite-time lifetime diverges as $\kappa/(2\epsilon^2)$. Exact hulls and fixed algebras describe the ideal endpoint; finite-time risks describe what remains available on the way to it. This is also the distinction needed by Foundation's hulls and the maintenance tasks of T-DOME and Brain Anchor. Neither the general stability estimate nor this example asserts that every slow operator space is an algebra or a physical projection.

# 5 Records live on the centre

## 5.1 A recording model

To see what the environment learns we replace the Markovian generator by an exact unitary model. Environmental fragments $E_1, \dots, E_K$ start in states $\omega_k$, and the interaction is

$$U = \sum_{x \in \mathcal{X}} |x\rangle\langle x| \otimes V_1(x) \otimes \cdots \otimes V_K(x), \tag{9}$$

with unitaries $V_k(x)$ that depend on the basis state through the coupling, for example $V_k(x) = \exp[-i\theta_k a_k(x) G_k]$ for a fragment generator $G_k$. The joint state after the interaction is

$$\sigma = \sum_{x,y} \rho_{xy} |x\rangle\langle y| \otimes \bigotimes_k V_k(x)\, \omega_k\, V_k(y)^\dagger,$$

so the system undergoes the Schur multiplier

$$\rho_{xy} \mapsto c(x,y)\, \rho_{xy}, \qquad c(x,y) = \prod_k c_k(x,y), \qquad c_k(x,y) = \mathrm{Tr}\big[\omega_k V_k(y)^\dagger V_k(x)\big]. \tag{10}$$

The generator (1) arises from such models in weak-coupling and repeated-collision limits. Here the collision statements are exact for the specified unitary and an initial product $\rho \otimes \bigotimes_k \omega_k$. Fresh independent fragments are an input: recollisions or shared initial fragment correlations do not obey the product formula automatically.

**Proposition 5.1** (Records carry only the block label)**.** *Let $\pi$ be a partition of $\mathcal{X}$ such that every $V_k(x)$ depends on $x$ only through its class $a(x) \in \pi$. Then:*

1. *$c(x,y) = 1$ whenever $x$ and $y$ lie in one class, so $\mathbb{E}_\pi(\rho)$, and with it the state on every block, is unchanged.*
2. *For every $O \in \mathfrak{B}_\pi$ and every operator $F$ on any collection $\mathcal{F}$ of fragments,*

$$\mathrm{Tr}[\sigma(O \otimes F)] = \sum_{a \in \pi} \mathrm{Tr}[P_a \rho P_a\, O]\ \mathrm{Tr}[\omega_{\mathcal{F}}(a) F], \qquad \omega_{\mathcal{F}}(a) = \bigotimes_{k \in \mathcal{F}} V_k(a)\, \omega_k\, V_k(a)^\dagger.$$

*The state of the fragments is $\sum_a p(a)\, \omega_{\mathcal{F}}(a)$ with $p(a) = \mathrm{Tr}\, P_a \rho$; it depends on $\rho$ only through the distribution of the label, and given the label, block observables and fragments are uncorrelated.*

*Proof.* (1) $V_k(x) = V_k(y)$ when $x, y$ lie in one class, so $c_k(x,y) = 1$. (2) $O_{yx} \neq 0$ only when $x$ and $y$ lie in one class $a$, and then the fragment factor is $\omega_{\mathcal{F}}(a)$ for both. Summing $\rho_{xy} O_{yx}$ over $x, y \in a$ gives $\mathrm{Tr}[P_a \rho P_a O]$. □

Whatever the environment learns under such couplings is the block label, which generates the centre of $\mathfrak{B}_\pi$. Many fragments can hold readable copies of that label while leaving the blocks alone. Block dependence alone does not guarantee distinguishability: the conditional states $\omega_k(a)$ must also be different, and a permitted measurement must resolve them. Records and the correlations inside the blocks are thus carried by one selected algebra, the first by its centre and the second by its blocks. That the redundantly available information is classical, and is information about one measurement, holds far more generally [18]; a two-qubit system with a decoherence-free subspace that proliferates part of its information while the remaining subspace keeps non-classical features was studied in [19]. What the commuting class adds is the exact location of both kinds of information in one algebra and, next, the price of imperfect symmetry.

## 5.2 Leakage costs coherence, and the costs multiply

**Theorem 5.2** (Leakage and coherence)**.** *For basis states $x, y$ and a fragment $k$, let $D_k(x,y) = \frac{1}{2}\big\| V_k(x)\omega_k V_k(x)^\dagger - V_k(y)\omega_k V_k(y)^\dagger \big\|_1$ be the fragment's distinguishability of $x$ and $y$, so that an optimal measurement on the fragment identifies one of the two equally likely states with probability $\frac{1}{2}(1 + D_k)$ [20]. Then*

$$|c_k(x,y)|^2 + D_k(x,y)^2 \leq 1, \tag{11}$$

*with equality when $\omega_k$ is pure. Consequently*

$$|c(x,y)| \leq \prod_k \big(1 - D_k(x,y)^2\big)^{1/2} \leq \exp\Big(-\tfrac{1}{2} \sum_k D_k(x,y)^2\Big), \tag{12}$$

*and if the coherence between $x$ and $y$ must stay above $1 - \varepsilon$, then*

$$\sum_k D_k(x,y)^2 \leq 2 \ln \frac{1}{1-\varepsilon}. \tag{13}$$

*Proof.* Purify $\omega_k = \mathrm{Tr}_R\,|\Omega\rangle\langle\Omega|$. Then $c_k(x,y) = \langle\Omega|(V_k(y)^\dagger V_k(x)\otimes I)|\Omega\rangle$ is the overlap of the pure states $(V_k(x)\otimes I)|\Omega\rangle$ and $(V_k(y)\otimes I)|\Omega\rangle$, whose trace distance is $(1-|c_k|^2)^{1/2}$. Partial trace over $R$ does not increase trace distance, which gives (11), with equality when there is nothing to trace out. Inequality (12) follows from (10) and $\ln(1-u)\le -u$, and (13) by taking logarithms. □

Inequality (11) is Englert's relation between visibility and which-way distinguishability [21], applied to one fragment. It bounds what a fragment learns by the coherence it destroys, not the reverse: a fragment in a mixed state can destroy coherence while learning nothing, as in Paper E's maximally mixed memory example. Purity is one useful preparation resource; mixed fragments can also record when their conditional states differ. For many fragments it becomes the budget (13): the sum of their squared pairwise distinguishabilities is at most $2\ln(1/(1-\varepsilon))$ for a protected pair of basis states. This is not a bound on their total Shannon information about an arbitrary block.

**Corollary 5.3** (Redundancy against symmetry)**.** *Suppose that $R$ fragments each distinguish two states $x, y$ of one block with $D_k(x,y)\ge\beta$. Then $|c(x,y)|\le(1-\beta^2)^{R/2}$, and keeping the coherence above $1-\varepsilon$ requires*

$$\beta \le \Big(\frac{2\ln(1/(1-\varepsilon))}{R}\Big)^{1/2}. \tag{14}$$

*In a sequence of models with $R\to\infty$ and $D_k(x,y)=\beta_R=\lambda R^{-\alpha}$, $\alpha>0$, for the $R$ fragments that couple to the block, the coherence tends to zero if $\alpha<\frac12$. If the fragment states are pure and no other fragment distinguishes $x$ from $y$, it tends to $1$ for $\alpha>\frac12$ and to $e^{-\lambda^2/2}$ for $\alpha=\frac12$.*

*Proof.* The bound and (14) are (12) and (13). For pure fragments $|c(x,y)| = (1-\beta_R^2)^{R/2}$ by the equality case, and $\frac{R}{2}\ln(1-\lambda^2R^{-2\alpha})\to 0,\ -\lambda^2/2,\ -\infty$ for $\alpha>\frac12, =\frac12, <\frac12$. □

A coupling of the form (9) is block-constant only by symmetry: the fragments must respond identically to all states of a block. Real couplings break such a symmetry slightly, and each recording fragment then leaks a little. Corollary 5.3 says that objectivity by redundancy and protection of the blocks can coexist, but only if the symmetry error per fragment falls as $R^{-1/2}$. In the limit of many records this is a sharp threshold in the exponent $\alpha$. It is the closest analogue of a transition in this class. It is a statement about how a symmetry error scales with the number of records, and it is not a phase transition in the thermodynamic sense.

## 6 Sector records and protected correlations

A mutual-information geometry does not require entanglement. Even local dephasing preserves $\rho = \frac12|00\rangle\langle 00| + \frac12|11\rangle\langle 11|$, for which $I(1:2)=\ln 2$. With local $Z$ couplings every selected block is one-dimensional and every sector state is a product. Its entire edge comes from sector weights.

For a selected state $\bar\rho = \mathbb{E}_\pi(\rho) = \sum_k p_k\rho_k$, with $\rho_k = P_k\rho P_k/p_k$, retain the label in a classical register:

$$\Omega_{SK} = \sum_k p_k\rho_k\otimes|k\rangle\langle k|_K. \tag{15}$$

This extension is physically realized by a perfectly resolving block-constant recorder followed by measurement or dephasing of its label register. It is not the coherent source–memory state before that readout. If the actual fragment states overlap, $K$ is still a useful mathematical label, but it is not a freely readable record. Zero-weight sectors are omitted.

**Proposition 6.1** (The two contributions to correlation)**.** *For any supplied disjoint subsystems $A, B$ of $S$,*

$$I(A:B)_{\bar\rho} = \sum_k p_k I(A:B)_{\rho_k} + I(A:K)_\Omega + I(B:K)_\Omega - I(AB:K)_\Omega. \tag{16}$$

*The last three terms need not have a nonnegative sum.*

*Proof.* For $X = A, B, AB$, the classical–quantum entropy formula gives $I(X:K) = S(\bar\rho_X) - \sum_k p_k S((\rho_k)_X)$. Substitution in $S(A)+S(B)-S(AB)$ proves the identity. □

The first term locates correlations protected within sectors; it can include classical as well as quantum correlations. The correction describes what mixing sector-conditioned states does to those correlations. In the locally dephased example above it equals $\ln 2$. In the next recording example it is negative. Thus neither a noncommutative block nor a classical centre alone determines the geometry. The sector states, their weights, the site marginals and the dictionary all enter.

## 7 How protected relations are formed

A noise commutant specifies what can be stored. It does not determine whether that content was inherited or created. The distinction already invalidates a noise-only mereology rule.

### 7.1 Disconnected noise can preserve a shared Bell pair

Take four qubits with only $A_{12} = Z_1 + Z_2$ and $A_{34} = Z_3 + Z_4$. Each pair has a zero-magnetization sector with logical basis $|0_L\rangle = |01\rangle$, $|1_L\rangle = |10\rangle$. The state

$$|\Psi_L\rangle = \frac{|0_L\rangle_{12}|0_L\rangle_{34} + |1_L\rangle_{12}|1_L\rangle_{34}}{\sqrt{2}} \tag{17}$$

is annihilated by both noise operators and is fixed exactly by (1). Yet $I(12:34) = 2\ln 2$. The two noise-hypergraph components are disconnected. Applying a Hadamard to each qubit gives the same counterexample for the $X$ couplings of Section 8. Classical and quantum correlations can both cross disconnected noise components when they are present initially.

### 7.2 A collective recorder forms a flagged protected relation

Start two qubits in the physical product $|+_X\rangle|+_X\rangle$ and a fresh three-level memory in $|0\rangle$. Let $M = Z_1 + Z_2$ and let $P_+, P_0, P_-$ project onto its eigenvalues $2, 0, -2$. For the shift $S_3|k\rangle = |k+1 \bmod 3\rangle$, use

$$U_{\rm rec} = P_+ \otimes I + P_0 \otimes S_3 + P_- \otimes S_3^2. \tag{18}$$

This is a block-constant recording interaction, not a product of independent local readers. It has a finite Hamiltonian implementation: choose a Hermitian logarithm $G$ with $e^{-iG} = S_3$ and $\|G\| = 2\pi/3$, put $a(M) = (2I - M)/2$, and apply $H_{\rm rec} = (\hbar/t_g)a(M) \otimes G$ for a time $t_g > 0$. The same memory couples to both sites. Its preparation, shared physical port, interaction time, control and subsequent readout are resources.

The complete output is

$$|\Omega\rangle = \tfrac{1}{2}|00\rangle|0\rangle + \tfrac{1}{\sqrt{2}}|\Psi^+\rangle|1\rangle + \tfrac{1}{2}|11\rangle|2\rangle, \qquad |\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}. \tag{19}$$

A computational-basis measurement of the memory produces the classical–quantum state with probabilities $(1/4, 1/2, 1/4)$ and conditional source states $(|00\rangle\langle 00|, |\Psi^+\rangle\langle\Psi^+|, |11\rangle\langle 11|)$. All branches remain in the protocol. Flag 1, of probability $1/2$, certifies a Bell pair in the $M = 0$ protected sector; flags $0, 2$ certify product states.

If the flag is discarded, the source is instead

$$\rho_* = \tfrac{1}{4}|00\rangle\langle 00| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \tfrac{1}{4}|11\rangle\langle 11|, \qquad I(1:2)_{\rho_*} = \tfrac{1}{2}\ln 2. \tag{20}$$

It is separable. Indeed it is the uniform collective $Z$-rotation twirl of $|+_X, +_X\rangle\langle +_X, +_X|$, a mixture of product rotations. Its partial-transpose eigenvalues are $0, 1/4, 1/4, 1/2$. The unconditional source therefore gains a mutual-information edge without entanglement; separability does not imply that every correlation measure is classical. The retained flagged ensemble permits a conditional protected Bell use, with herald probability and every unsuccessful branch charged. The pure source–memory state (19), the measured flagged state and the unconditional source are three different outputs.

For this flagged ensemble, $\sum_k p_k I(1:2)_{\rho_k} = \ln 2$, $I(1:K) = I(2:K) = \frac{1}{2}\ln 2$ and $I(12:K) = H(K) = \frac{3}{2}\ln 2$. Equation (16) recovers the unconditional $\frac{1}{2}\ln 2$ with correction $-\frac{1}{2}\ln 2$. Collective dephasing of the product input alone reaches the same $\rho_*$ but supplies no accessible flag. A flag requires the specified recording interaction and actual readout.

### 7.3 A real link forms a deterministic protected Bell pair

The distinction is sharper in the four-qubit example. Define

$$L^x_{ab} = \tfrac{1}{2}(X_aX_b + Y_aY_b), \qquad H_{\rm link} = \hbar J L^x_{12}L^x_{34}, \quad J > 0.$$

$L^x_{ab}$ exchanges $|01\rangle$ and $|10\rangle$ and vanishes on $|00\rangle, |11\rangle$. Consequently $[H_{\rm link}, A_{12}] = [H_{\rm link}, A_{34}] = 0$. Starting from the physical product $|01\rangle_{12}|01\rangle_{34}$, the full noisy evolution remains in the joint zero sector and is exactly

$$|\psi(t)\rangle = \cos(Jt)|0_L0_L\rangle - i\sin(Jt)|1_L1_L\rangle. \tag{21}$$

At $t = \pi/(4J)$ it has $I(12:34) = 2\ln 2$ and one ebit across the pair partition. The noise term vanishes throughout the trajectory.

This is new protected correlation from an initially uncorrelated state. It uses an actual cross-pair Hamiltonian link, here a four-body control term. The noise graph remains disconnected but the complete interaction graph does not. The factorization, prepared input, gate permissions, coupling strength and duration are supplied; there is no inferred free implementation or heat price for the link.

### 7.4 The process constraint and learned use

**Proposition 7.1** (Correlation provenance across independent components)**.** *Let a supplied partition of a carrier have initially independent states and auxiliary resources. Suppose its full channel, including record registers, factors across these components. Then it creates no cross-component mutual information. If the input is correlated, a product channel cannot increase the mutual information across any bipartition of the components, but it can preserve initially shared classical or quantum correlations.*

*Proof.* A product channel sends a product input to a product output, including its local records. For a correlated input, $I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, so the second statement is data processing under the product channel. □

An implementation satisfies the factor premise when its Hamiltonian and noise generators act separately on the components, its fresh ports are independent, and every control and instrument is component-local. Shared random controls, communication or common memories must be added to the process description: they can correlate unconditional outputs. Conditioning on a joint event can also change correlations and is not a product unconditional channel. The proposition is a diagnostic, not a claim that every process with a disconnected noise graph factors.

The constructions above do more than add initial-state qualifications to a conjecture. They exhibit how the missing relations can form: a collective port records an indistinguishability sector, while an interaction inside protected sectors builds usable quantum content. The other indispensable step is learning which encoding an unknown physical plant protects. Foundation III [6] implements that step for $A_s = n_1 + sn_2$ with unknown $s = \pm 1$. A diagnostic sharing the plant's actual sign writes a record; that record, not $s$, selects odd or even encoding and the matching decoder. The same logical content and its old reference survive from before diagnosis. The record-controlled frame changes the later contact, rather than merely changing a description of the fixed commutant.

That result assumes a physical relation between diagnostic and storage couplings. A diagnostic carrying an unrelated sign would not learn the protected frame. Its retained-history scalar/JT implementation also contains real body forces, a source and a second acquisition; it is not identified with the full quantum material model. The complementary contribution here is the separation of formation, protection and use: each can fail while the others hold. A noiseless block can be unreadable, a readable label can outlive its quantum content, and a learned encoder can protect content without supplying every logical gate.

## 8 A geometry selected by the coupling

We now apply these results to the state and the reconstruction rule of Paper A. The rule acts on the complete graph of the sites: the edge $\{i,j\}$ has length $1/I_{ij}$, where $I_{ij}$ is the mutual information of sites $i$ and $j$, and distances are shortest-path lengths. We extend it to $I_{ij} = 0$ by giving that edge infinite length, so that a site whose edges all have zero mutual information is disconnected. The state is the three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$. In the eigenbasis $|\pm\rangle$ of the Pauli operator $X$ it reads

$$|\mathrm{GHZ}\rangle = \tfrac{1}{2}\big(|{+}{+}{+}\rangle + |{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle\big).$$

All couplings below are sums of $X_i$, hence diagonal in this basis, and Sections 3–5 apply. A local coupling $A = X_i$ with rate $\gamma_i$ is the bit-flip noise of Paper A, with flip probability $p_i = (1 - e^{-2\gamma_i t})/2$. A collective coupling $A = X_1 + X_2 + X_3$ dephases between different values of the $X$-magnetization $M = \sum_i x_i$ and leaves states of equal $M$ coherent.

**Proposition 8.1** (Three couplings, three geometries)**.** *Let* $\Psi^+ = (|{+}{-}\rangle + |{-}{+}\rangle)/\sqrt{2}$ *and* $\tilde{W} = (|{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle)/\sqrt{3}$*. The long-time selected states* $\mathbb{E}_{\pi_\infty}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|)$ *and their geometries are:*

1. *Collective coupling* $X_1 + X_2 + X_3$*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{3}{4}|\tilde{W}\rangle\langle\tilde{W}|$*. Each pair has* $I_{ij} = \frac{1}{2}\ln 2$*; the geometry is an equilateral triangle with side* $2/\ln 2 = 2.8854$*.*

2. *Collective coupling $X_1 + X_2$ and local coupling $X_3$:* $\frac{1}{4}|++\rangle\langle+++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+|_{12} \otimes |-\rangle\langle-|_3 + \frac{1}{4}|--+\rangle\langle--+|$. *Here $I_{12} = \frac{1}{2}\ln 2$ and $I_{13} = I_{23} = 0$; the geometry is a single edge, and site 3 is disconnected.*

3. *Local couplings $X_1, X_2, X_3$: the uniform mixture of the four even strings $|+++\rangle, |+--\rangle, |-+-\rangle, |--+\rangle$. Every $I_{ij} = 0$; no edge survives.*

*Proof.* The pinching keeps the amplitudes inside each class of equal eigenvalue tuples. (1) The classes met by the GHZ state are $M = 3$ and $M = -1$. The marginal of sites $1, 2$ is $\frac{1}{4}|++\rangle\langle++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \frac{1}{4}|--\rangle\langle--|$, with eigenvalues $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and entropy $\frac{3}{2}\ln 2$; each single-site marginal is $I/2$; so $I_{12} = 2\ln 2 - \frac{3}{2}\ln 2$, and the other pairs follow by symmetry. (2) The classes are labelled by $(x_1 + x_2, x_3)$; the strings $|+--\rangle$ and $|-+-\rangle$ share the label $(0, -)$. The marginal of sites $1, 2$ is as in (1). The marginal of sites $1, 3$ is uniform on the four strings $|++\rangle, |+-\rangle, |--\rangle, |-+\rangle$, and so is that of sites $2, 3$. (3) Every pair marginal of the uniform mixture of even strings is uniform. Direct evaluation of the full eight-dimensional density matrices gives the same values. □

The state and the rule are the same in all three cases; only the coupling differs. At zero time each pair has $I_{ij} = \ln 2$, Paper A's GHZ geometry at zero noise. Under local noise Paper A's triangles relax toward the empty graph; a collective coupling stops the relaxation at an equilateral triangle, and a coupling that is collective on one pair keeps exactly that edge. The surviving correlations come from blocks of dimension greater than one: the three-dimensional block $M = -1$, which contains the entangled state $\tilde{W}$, in case (1), and the two-dimensional block spanned by $|+--\rangle$ and $|-+-\rangle$ in case (2). In case (3) the selected algebra is commutative with product minimal projections, every selected state is classically correlated, and for this state the classical pair correlations vanish as well. By Proposition 3.1(4) the channels at any finite time generate the full matrix algebra in all three cases, so an algebra label cannot tell them apart.

### 8.1 The lifetime of the selected geometry

Real couplings are not exactly collective. Add weak local couplings $X_i$ with rates $\gamma_i$ to the collective coupling with rate $\gamma_c$. The exact fixed algebra is then commutative and the long-time geometry is empty, but on a window the magnetization blocks are still selected. With $\pi$ the partition by $M$, the rates (3) are $\Gamma_{\text{in}}(\pi) = 2(\gamma_1 + \gamma_2 + \gamma_3)$ and $\Gamma_{\text{out}}(\pi) = 2\gamma_c + 2\min_i \gamma_i$, while the coherences of the GHZ state between its two blocks decay at the rates $8\gamma_c + 2(\gamma_i + \gamma_j)$.

**Lemma 8.2** (Plateau bound)**.** *For a state $\rho$ and a partition $\pi$, let $\Gamma_{\text{out}}(\rho, \pi)$ be the smallest $D(x, y)$ over pairs in different classes with $\rho_{xy} \neq 0$, and let $r$ be the number of basis vectors with $\rho_{xx} \neq 0$. Then for all $t \geq 0$*

$$\|\Phi_t(\rho) - \mathbb{E}_\pi(\rho)\|_1 \leq \sqrt{r}\, e^{-t\Gamma_{\text{out}}(\rho,\pi)}\, \|\rho - \mathbb{E}_\pi(\rho)\|_2 + t\,\Gamma_{\text{in}}(\pi). \tag{22}$$

*Proof.* Split $\Phi_t(\rho) - \mathbb{E}_\pi\rho = \Phi_t(\rho - \mathbb{E}_\pi\rho) + (\Phi_t(\mathbb{E}_\pi\rho) - \mathbb{E}_\pi\rho)$. The second term is bounded by Theorem 4.2(1). The first is the Schur product of $\rho - \mathbb{E}_\pi\rho$ with multipliers of modulus at most $e^{-t\Gamma_{\text{out}}(\rho,\pi)}$; its rank is at most $r$, because a row of a positive matrix with zero diagonal entry vanishes, and $\|Z\|_1 \leq \sqrt{\operatorname{rank} Z}\, \|Z\|_2$. □

The bound turns into a certified geometry through continuity of entropy. If two states of a qubit pair are at trace distance $\delta = \frac{1}{2}\|\rho - \sigma\|_1 \leq \frac{1}{2}$, then by the Fannes–Audenaert inequality [22] and Winter's bound for conditional entropy [23] their mutual informations differ by at most

$$f(\delta) = h(\delta) + 2\delta\ln 2 + (1 + \delta)\, h\Big(\frac{\delta}{1 + \delta}\Big), \tag{23}$$

with $h$ the binary entropy in nats. Partial traces do not increase trace distance, and Paper A's stability bound for graph metrics converts a uniform change $f < m$ of mutual informations bounded below by $m$ into a change of at most $2f/(m(m - f))$ of every distance among three sites.

**Proposition 8.3** (A certified plateau)**.** *For $\gamma_c = 1$ and $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$, the bounds* (22) *and* (23) *certify that every distance of the GHZ geometry lies within $25\%$ of the plateau value $2/\ln 2$ for $0.716 \leq t \leq 3.58$. Numerical root finding of the exact evolution gives the approximate $25\%$ window $0.053 \lesssim t \lesssim 53.6$ and $5\%$ window $0.163 \lesssim t \lesssim 8.86$. If the local rates are scaled to $s(1, 2, 4)$ with $s \leq 10^{-3}$, the late edges scale to leading order as $1/s$: approximately $5.36 \times 10^{-3}/s$ for the $25\%$ window, $8.86 \times 10^{-4}/s$ for the $5\%$ window and, when $s \leq 10^{-4}$ so that it exists, $3.58 \times 10^{-4}/s$ for the certified $25\%$ window. The windows begin at times of order $1/\gamma_c$.*

*Proof.* For this state $r = 4$, $\|\rho - \mathbb{E}_\pi\rho\|_2 = \sqrt{6}/4$, $\Gamma_{\text{out}}(\rho, \pi) = 8.0006$ and $\Gamma_{\text{in}}(\pi) = 1.4 \times 10^{-3}$. The right side of (22) is the convex function $b(t) = (\sqrt{6}/2)e^{-8.0006t} + 0.0014t$. On $[0.716, 3.58]$ its maximum is at an endpoint and is

| $t$ | $\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1$ | bound (22) | distances $(d_{12}, d_{13}, d_{23})$ |
|---|---|---|---|
| 0 | 0.8660 | 1.2247 | (1.4427, 1.4427, 1.4427) |
| 1 | 0.0010 | 0.0018 | (2.8968, 2.9034, 2.9066) |
| 10 | 0.0094 | 0.0140 | (2.9728, 3.0221, 3.0458) |
| 100 | 0.0897 | 0.1400 | (3.5620, 3.9855, 4.2034) |
| 300 | 0.2442 | 0.4200 | (4.8943, 6.5032, 7.4508) |
| 1000 | 0.5969 | 1.4000 | (12.559, 28.867, 41.426) |

Table 1: GHZ state under a collective coupling with $\gamma_c = 1$ and local couplings with $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$. The plateau value of every distance is $2/\ln 2 = 2.8854$.

less than $0.0050121$. The trace distance used in (23) is $\delta(t) = \frac{1}{2}\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1 \le b(t)/2$, not $b(t)$. The increasing entropy and graph bounds therefore give a distance error less than $0.72130 < \frac{1}{4} \cdot 2/\ln 2$ throughout that interval. The exact-evolution window edges were located by root finding of the Schur multipliers. Once the coherences between blocks have decayed, the in-block multipliers depend on $t$ only through $st$, which gives the late-edge scaling. The root values are numerical diagnostics, distinct from the conservative analytic interval. □

Table 1 shows the three stages. The GHZ geometry, with side $1/\ln 2$, relaxes on the time scale $1/\gamma_c$ to the selected triangle with side $2/\ln 2$; the triangle persists while $t\Gamma_{\text{in}}(\pi)$ is small; then the unequal local rates deform it into scalene triangles, which are not isometric to the plateau, before every edge diverges. This is Paper A's non-isometry produced by one time-ordered dynamics rather than by a comparison of separately chosen channels. Proposition 8.3 gives a state-specific stability certificate for Paper A's declared graph dictionary on an explicit window. The plateau exists when its end lies after its beginning, that is, when the ratio of in-block to between-block rates is small, as Theorem 4.2 requires.

The constants in Proposition 8.3 are small. At the end of the exact $25\%$ window the three in-block coherences have fallen by only $3.2$, $5.2$ and $6.2$ per cent. The selected pair marginal has a zero eigenvalue, and in-block decoherence populates it, so pair entropies respond to a coherence loss $\delta$ like $\delta\ln(1/\delta)$. Geometries read from entropies are therefore far more fragile than the coherences that carry them, and the gap between the certified and the exact window comes mostly from this logarithmic response together with worst-case continuity bounds.

### 8.2 Imperfect redundant records detach a vertex

Finally we let fragments record the magnetization with a small asymmetry. Each of $K$ fragment qubits starts in $|0\rangle$ and interacts through $V_k(x) = \exp[-i\theta(M(x) + \eta x_1)Y]$ with $\theta = \pi/8$, where $Y$ is the Pauli operator of the fragment. For $\eta = 0$ the coupling is block-constant. The two magnetizations present in the GHZ state, $M = 3$ and $M = -1$, then produce orthogonal fragment states, so every fragment holds a perfect copy of the label, and by Proposition 5.1 the equilateral triangle survives for any number of fragments. For $\eta > 0$ each fragment distinguishes strings of the block $M = -1$ that differ at site 1, with $D_k = |\sin 2\theta\eta|$.

| $\eta$ | $K$ | coherence of $\|+--\rangle, \|-+-\rangle$ | $I_{12} = I_{13}$ | $I_{23}$ | label error |
|---|---|---|---|---|---|
| 0.02 | 1 | 0.99988 | 0.34631 | 0.34657 | $6.2 \times 10^{-5}$ |
| 0.02 | 100 | 0.98774 | 0.32790 | 0.34657 | $6.2 \times 10^{-5}$ |
| 0.02 | 1000 | 0.88393 | 0.23581 | 0.34657 | $6.2 \times 10^{-5}$ |
| 0.05 | 1 | 0.99923 | 0.34525 | 0.34657 | $3.9 \times 10^{-4}$ |
| 0.05 | 100 | 0.92577 | 0.26724 | 0.34657 | $3.9 \times 10^{-4}$ |
| 0.05 | 1000 | 0.46243 | 0.05555 | 0.34657 | $3.9 \times 10^{-4}$ |

Table 2: Recording the magnetization with a site-1 asymmetry $\eta$. The last column is the minimal error probability with which a measurement on one fragment identifies the label $M$, whose prior is $(\frac{1}{4}, \frac{3}{4})$. Coherences are relative to their initial values; the coherence between $|-+-\rangle$ and $|--+\rangle$ stays equal to 1.

Table 2 shows the effect. Every fragment is a nearly perfect and readable record of the label for both asymmetries, so the loss of geometry is not a price paid for better records. It is the small leakage about site 1, multiplied over fragments as in Theorem 5.2. The coherences between strings that differ at site 1 decay as $(1 - D_k^2)^{K/2}$, the edges to site 1 weaken, and as $K \to \infty$ the state approaches $\frac{1}{4}|+++\rangle\langle+++| + \frac{1}{4}|+--\rangle\langle+--| + \frac{1}{2}|-\rangle\langle-|_1 \otimes |\Psi^+\rangle\langle\Psi^+|_{23}$. Then

$I_{12} = I_{13} = 0$ and $I_{23} = \frac{1}{2}\ln 2$: site 1 is detached and the geometry becomes the single edge of Proposition 8.1(2), now on sites $2, 3$. With $\eta_K = 0.5K^{-\alpha}$ the coherence after $K = 10^6$ fragments is 0 for $\alpha = \frac{1}{4}$, $0.9258 = e^{-(\pi/8)^2/2}$ for $\alpha = \frac{1}{2}$, and $0.9999$ for $\alpha = \frac{3}{4}$, as Corollary 5.3 predicts with $\lambda = 2\theta \cdot 0.5$ to leading order.

## 9 What selection settles

The selected partition is fixed by the supplied dynamics, window and tolerance, and is robust to perturbations preserving the rate gap. This differs in emphasis from QBism's state assignments and relational quantum mechanics' reference-system dependence [26, 25], without deciding their wider interpretations. Public agreement further needs actual fragments and calibrated readers; Paper A provides one finite realization. A stable algebra alone does not grant those permissions.

The GHZ comparison changes a state under one graph dictionary, not the coordinates of an unchanged metric. Sector mixing can sustain edges after quantum content is lost; the leaking GHZ block instead supports the specific edges destroyed in Table 2. Formation requires the collective port or Hamiltonian link of Section 7, not noise selection alone. Removing either resource supplies its corresponding opposing control.

AQFT and Consciousness I start with their local or current-net carriers [24, 4]. Observable-induced factorization and quantum mereology address when retained relations support an effective subsystem description [28, 29, 30]. Decoherence and Everettian accounts share the underlying dynamics [10, 27]; the issue here is whether its resulting structure is readable, operable and persistent for a finite task.

## 10 Beyond the commuting class

Conjecture 2.1 separates the existence of a stable block from its acquisition as an operational component. We now examine the further conditions required outside the exactly commuting class.

The conjecture concerns effective factorization, not the existence of elementary sites. It predicts more than correlation alone: successful formation should yield an encoder, decoder and repeatable operations on the formed component, with calibration records and finite error bounds. The exact formation examples give protected relational content; Foundation III adds record-controlled encoding of unknown content. Together they support a route from coupling through history to changed capability. They do not yet supply a general autonomous acquisition of a complete internal gate set. A proposed component that has only a state-dependent correlation but no reproducible internal operations would fail this criterion. The disconnected Bell counterexample rules out noise-only connectivity as a necessary condition, while Proposition 7.1 identifies the real no-creation obstruction for product processes.

For general Lindblad dynamics, a separated spectrum defines slow states and an effective metastable evolution [17]. A finite-window extension should select the operational structure of this manifold rather than assume that slow observables close under multiplication. Even complete positivity and idempotence do not ensure closure in the original multiplication. For example,

$$P(X) = \operatorname{diag}\big(x_{11}, x_{22}, (x_{11} + x_{22})/2\big) \tag{24}$$

is a unital completely positive idempotent on $M_3$. It is completely positive because it is a sum of two positive evaluation maps into positive diagonal matrices. Its range contains $E = \operatorname{diag}(1, 0, 1/2)$ but not $E^2 = \operatorname{diag}(1, 0, 1/4)$. A completely positive projection gives its range a Choi–Effros product $x \circ y = P(xy)$ [7, 8]. This abstract product does not make $xy$ a physically permitted observable or control in the original representation. An ordinary conditional expectation onto a represented subalgebra needs the additional algebraic closure. A metastable spectral projection may not even be completely positive. Thus a general selection theory must provide actual instruments or recovery maps, their errors and their operational closure, rather than infer them from a slow spectral space.

The physical price of that capability is the next link to Q-RAIF and the gravitational line. A passive GHZ geometry has lifetime of order $1/s$ under symmetry breaking, while independent redundant fragments require an $R^{-1/2}$ protecting symmetry. Q-RAIF B realizes correction as field coupling with work and stress [5]; HAFF D distinguishes entropy removal, usable side information and supplied purity, and gives specified source–response constructions [2]. These results suggest a maintained-structure mechanism: actual records learn a protected frame, operations install it, maintenance changes the physical source, and the response changes subsequent records. Foundation III realizes a finite retained-history scalar/JT return, whereas the full quantum material source requires its own support and preparation account. Connecting those implementations and renewing their resources is a substantive requirement, not an irreducible heat flux inferred from algebra inclusion.

Many-site and continuum selection require further work when rate gaps close. Consciousness I's ideal fixed-net closure and finite-time noise estimates are not already a continuum uniqueness theorem for window-selected blocks. Likewise, the graph dictionary is supplied here. A general emergence of spacetime needs a physical family of readers and a reconstruction law that remains controlled as the operational structure grows.

## 11 Conclusion

A commuting coupling determines its fixed block algebra and its long-time pinching. On a separated finite window it selects a unique block partition in the supplied joint eigenbasis; the selected algebra contains every stable block algebra of that form. The actual finite-time channel has full operator range. Block-constant record couplings report sector weights without resolving protected content. Independent leakage multiplies the coherence loss and imposes a sharp redundancy–symmetry threshold.

The relation to geometry has two parts. Sector mixing and within-sector content contribute through the exact identity (16); neither alone determines the mutual information graph. For the GHZ source, collective, pair-collective and local couplings yield a triangle, one edge and no edge under one rule. Weak asymmetry gives a finite plateau and then destroys its supporting edges. Noncommuting perturbations can erase the exact fixed algebra without abruptly erasing a finite record or task capability.

Formation requires a history. An initially shared logical Bell pair can survive between noise-disconnected protected components, so a noise graph cannot delimit all surviving quantum relations. A collective recorder forms a flagged protected Bell branch from a product input, with all branches retained; an actual Hamiltonian link forms a deterministic Bell pair protected throughout its generation. These constructions reduce the missing formation step to identifiable physical operations without hiding their costs. Record-controlled encoding supplies the further route from learned relations to changed capability. The remaining general problem is to derive reproducible subsystem operations and their sustained maintenance from such histories, rather than merely rename correlated states as subsystems.

## References


[1] S. Liu, *Physical Coarse-Graining and the Problem of Emergent Geometry: The Holographic Alaya-Field Framework*, 2026; series record DOI: 10.5281/zenodo.18361706.

[2] S. Liu, *Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return*, (2026), series record DOI: 10.5281/zenodo.18388881.

[3] S. Liu, *Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout*, (2026), series record DOI: 10.5281/zenodo.18400065.

[4] S. Liu, *Geometry Relative to a Cut: Selection, Generation and Modular Inseparability*, (2026), series record DOI: 10.5281/zenodo.19607387.

[5] S. Liu, *Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost*, (2026), series record DOI: 10.5281/zenodo.18525890.

[6] S. Liu, *Foundation III: The Perfuming Loop. Held Content, Learned Frames and Physical Return*, 2026, in the Genesis Trilogy working corpus.

[7] M.-D. Choi and E. G. Effros, *Injectivity and operator spaces*, J. Funct. Anal. **24**, 156–209 (1977).

[8] D. P. Blecher and M. Neal, *Completely contractive projections on operator algebras*, Pacific J. Math. **283**, 289–324 (2016).

[9] W. H. Zurek, *Environment-induced superselection rules*, Phys. Rev. D **26**, 1862 (1982).

[10] W. H. Zurek, *Decoherence, einselection, and the quantum origins of the classical*, Rev. Mod. Phys. **75**, 715 (2003).

[11] W. H. Zurek, *Quantum Darwinism*, Nature Physics **5**, 181 (2009).

[12] P. Zanardi and M. Rasetti, *Noiseless quantum codes*, Phys. Rev. Lett. **79**, 3306 (1997).

[13] E. Knill, R. Laflamme, and L. Viola, *Theory of quantum error correction for general noise*, Phys. Rev. Lett. **84**, 2525 (2000).

[14] A. Frigerio, *Stationary states of quantum dynamical semigroups*, Commun. Math. Phys. **63**, 269 (1978).

[15] R. Blume-Kohout, H. K. Ng, D. Poulin, and L. Viola, *Information-preserving structures: A general framework for quantum zero-error information*, Phys. Rev. A **82**, 062306 (2010).

[16] P. Facchi and S. Pascazio, *Quantum Zeno subspaces*, Phys. Rev. Lett. **89**, 080401 (2002).

[17] K. Macieszczak, M. Guţă, I. Lesanovsky, and J. P. Garrahan, *Towards a theory of metastability in open quantum dynamics*, Phys. Rev. Lett. **116**, 240404 (2016).

[18] F. G. S. L. Brandão, M. Piani, and P. Horodecki, *Generic emergence of classical features in quantum Darwinism*, Nature Commun. **6**, 7908 (2015).

[19] B. Çakmak, Ö. E. Müstecaplıoğlu, M. Paternostro, B. Vacchini, and S. Campbell, *Quantum Darwinism in a composite system: Objectivity versus classicality*, Entropy **23**, 995 (2021).

[20] C. W. Helstrom, *Quantum Detection and Estimation Theory*, Academic Press (1976).

[21] B.-G. Englert, *Fringe visibility and which-way information: An inequality*, Phys. Rev. Lett. **77**, 2154 (1996).

[22] K. M. R. Audenaert, *A sharp continuity estimate for the von Neumann entropy*, J. Phys. A **40**, 8127 (2007).

[23] A. Winter, *Tight uniform continuity bounds for quantum entropies: Conditional entropy, relative entropy distance and energy constraints*, Commun. Math. Phys. **347**, 291 (2016).

[24] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, Springer-Verlag (1996).

[25] C. Rovelli, *Relational quantum mechanics*, Int. J. Theor. Phys. **35**, 1637 (1996).

[26] C. A. Fuchs, N. D. Mermin, and R. Schack, *An introduction to QBism with an application to the locality of quantum mechanics*, Am. J. Phys. **82**, 749 (2014).

[27] D. Wallace, *The Emergent Multiverse*, Oxford University Press (2012).

[28] P. Zanardi, D. A. Lidar, and S. Lloyd, *Quantum tensor product structures are observable induced*, Phys. Rev. Lett. **92**, 060402 (2004).

[29] J. Cotler, G. Penington, and D. H. Ranard, *Locality from the spectrum*, Commun. Math. Phys. **368**, 1267 (2019).

[30] S. M. Carroll and A. Singh, *Quantum mereology: Factorizing Hilbert space into subsystems with quasiclassical dynamics*, Phys. Rev. A **103**, 022213 (2021).